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UNIT 2 ELECTRICAL MACHINES AND MEASUREMENTS			
Q.No.	**Question**	**BTL**	
1.	A transformer in a power supply has a primary voltage of **230 V** and a secondary voltage of **12 V**. If the secondary current is **2 A**, what is the approximate primary current (neglecting losses)? A) 0.1 A B) 0.2 A C) 0.1 A D) 0.104 A Answer. D. 0.104 A	3	Entered
2.	A **4-pole DC generator** has **wave wound armature** with **400 conductors**. The flux per pole is **0.02 Wb** and it runs at **1500 rpm**. Find the **generated EMF**. A) 400 V B) 800 V C) 200 V D) 600 V Answer. A. 400 V	3	Entered
3.	A **DC shunt motor** runs at **1000 rpm** and takes **5 A** at **220 V**. The armature resistance is **0.5 Ω**, and the field current is **1 A**. Find the **back EMF (Eb)**. A) 215 V B) 217.5 V C) 210 V D) 200 V Answer. B. 217.5 V	3	Entered
4.	A **DC shunt motor** has back EMF **E_1 = 200 V** at speed **1000 rpm**. When the speed is reduced to **800 rpm**, flux increases by **10%**. Find the **new back EMF**. A) 160 V B) 176 V C) 180 V D) 200 V Answer. B. 176 V	3	Entered

5.	A DC shunt generator supplies 220 V to a load drawing 20 A. The armature resistance is 0.5 Ω. Find the generated EMF (E_g). A) 220 V B) 225 V C) 230 V D) 240 V Answer. B. 225 V	4	Entered
6.	A DC motor runs at 1200 rpm when drawing 40 A from a 250 V supply. The armature resistance is 0.4 Ω. Find the back EMF. A) 240 V B) 234 V C) 236 V D) 238 V Answer. B. 234 V	4	Entered
7.	A DC motor develops a torque of 50 N·m when armature current is 25 A. Find the armature torque constant (K_T). A) 1 N·m/A B) 2 N·m/A C) 3 N·m/A D) 4 N·m/A Answer. B. 2 N·m/A	4	Entered
8.	A DC generator delivers 10 kW at 250 V. Armature resistance = 0.2 Ω, shunt field resistance = 250 Ω. Find the generated EMF (Eg). A) 250 V B) 254 V C) 255 V D) 260 V Answer. C. 255 V	5	Entered
9.	A DC shunt motor draws 10 A from 220 V supply, with R_a = 0.5 Ω, R_f = 220 Ω. Find the mechanical power developed. A) 2100 W B) 2150 W C) 2189 W D) 2200 W Answer. C. 2189 W	5	Entered
10.	A DC motor runs at 1000 rpm with a back EMF of 200 V. If armature current and flux remain constant, what will be the speed when back EMF becomes 180 V?4 A) 800 rpm B) 850 rpm	5	Entered

	C) 900 rpm D) 950 rpm Answer. C. 900 rpm		
11	A DC series motor has full-load current of 30 A at 250 V. If the armature and series field resistances together are 0.5 Ω, find the back EMF at full load. A) 235 V B) 240 V C) 245 V D) 250 V Answer. A. 235 V	3	Entered
12	A DC shunt motor runs at 1200 rpm with a back EMF of 220 V. When the field current is reduced by 10%, the flux decreases by 10%, and armature current remains constant. Find the new speed. A) 1200 rpm B) 1333 rpm C) 1100 rpm D) 1000 rpm Answer. B. 1333 rpm	3	Entered
13	A single-phase induction motor has a synchronous speed of 1500 rpm and an actual speed of 1440 rpm. Find the slip. A) 2% B) 3% C) 4% D) 5% Answer. C. 4%	3	Entered
14	If the frequency of the supply to a 4-pole single-phase induction motor is 50 Hz, find the synchronous speed. A) 750 rpm B) 1000 rpm C) 1500 rpm D) 3000 rpm Answer. C. 1500 rpm	4	Entered
15	A single-phase induction motor takes 8 A at 230 V and delivers 1 kW output power. Find its efficiency. A) 50% B) 54% C) 60% D) 65% Answer. D. 65%	5	Entered
16	A single-phase induction motor has rotor copper loss = 60 W, stator copper loss = 40 W, and core loss = 30 W. If input power = 1500 W, find mechanical output power. A) 1350 W B) 1370 W C) 1375 W D) 1390 W Answer. B. 1370 W	3	Entered

17	A single-phase induction motor running at 0.05 slip develops 1 kW rotor input power. Find the rotor copper loss. A) 10 W B) 25 W C) 50 W D) 100 W Answer. C. 50 W	4	Entered
18	If a single-phase induction motor has a rotor input = 1000 W and operates at a slip of 4%, find the mechanical power developed. A) 960 W B) 940 W C) 900 W D) 920 W Answer. A. 960 W	5	Entered
19	A single-phase induction motor runs at 1470 rpm on a 50 Hz supply. Calculate the rotor frequency if it has 4 poles. A) 1.5 Hz B) 2.0 Hz C) 2.5 Hz D) 3.0 Hz Answer. B. 2.0 Hz	3	Entered
20	A 240 V, 50 Hz, 4-pole single-phase induction motor runs at 1450 rpm and draws 5 A. Calculate the input mechanical power assuming PF = 0.8 and efficiency $\eta = 0.7$. A) 500 W B) 672 W C) 672 W D) 750 W Answer. B. 672 W	3	Entered
21	A single-phase induction motor has a synchronous speed of 1500 rpm. If its rotor speed is 1425 rpm, find the rotor copper loss per rotor input. A) 4% B) 5% C) 6% D) 7% Answer. A. 4%	3	Entered
22	A 240 V single-phase induction motor draws 6 A at 0.7 PF lag. If the output is 800 W, find the efficiency. A) 40% B) 50% C) 60% D) 70% Answer. C. 60%	3	Entered
23	For a single-phase induction motor, the starting torque (T_{st}) is 2.5 N·m and the full-load torque (T_{fl}) is 4 N·m. Find the starting torque ratio.	4	Entered

	A) 0.25 B) 0.5 C) 0.625 D) 1.5 Answer. C. 0.625		
24	If a single-phase induction motor operates at slip = 0.03, and the rotor input = 600 W, find the mechanical power developed and rotor Cu loss. A) 582 W, 18 W B) 594 W, 6 W C) 580 W, 20 W D) 570 W, 30 W Answer. A.	4	Entered
25	A single-phase induction motor develops mechanical output = 300 W, slip = 0.04, and efficiency = 75%. Find the stator input power. A) 380 W B) 400 W C) 420 W D) 450 W Answer. C. 420 W	5	Entered
26	A split-phase induction motor has a starting winding resistance of 10 Ω and reactance of 20 Ω. Find the phase angle between current and voltage in the starting winding. A) 26.6° B) 45° C) 63.4° D) 30° Answer. C. 63.4°	5	Entered
27	In a split-phase induction motor, the main winding has impedance $Z_m=25\angle 60°$ Ω and the auxiliary winding has $Za=15\angle 30°$ Ω. Find the phase difference between the two currents. A) 30° B) 45° C) 60° D) 90° Answer. B. 45°	4	Entered
28	A capacitor-start induction motor has a main winding current of 4 A (lagging supply by 40°) and auxiliary winding current of 3 A (leading supply by 50°). Find the phase angle between the two currents. A) 80° B) 85° C) 90° D) 95° Answer. D. 95°	5	Entered
29	A capacitor-start single-phase induction motor has main winding current 4 A and auxiliary winding current 3 A with phase difference 80°. Find the starting torque proportionality factor.	4	Entered

	A) 11.7 B) 12.0 C) 12.5 D) 14.0 Answer. B. 12.0		
30	A capacitor-start motor operates on a 240 V, 50 Hz supply with a starting capacitor of 120 μF and auxiliary winding reactance 30 Ω. Find the reactive voltage across the capacitor at starting. A) 100 V B) 200 V C) 300 V D) 400 V Answer. B. 200 V	3	Entered
31	A capacitor-start and run motor uses two capacitors, one for starting (C_1) and one for running (C_2). If C_1=150μF and C_2=25μF, find the equivalent capacitance during starting. A) 150 μF B) 175 μF C) 125 μF D) 200 μF Answer. B. 175 μF	3	Entered
32	A capacitor-run motor with a running capacitor of 25 μF operates at 50 Hz. Find its capacitive reactance. A) 106 Ω B) 127 Ω C) 159 Ω D) 178 Ω Answer. C. 159 Ω	3	Entered
33	A capacitor-start induction motor develops a starting torque 1.5 times the full-load torque. If full-load torque = 3 N·m, find the starting torque. A) 3 N·m B) 4.5 N·m C) 5 N·m D) 6 N·m Answer. B. 4.5 N·m	3	Entered
34	In a capacitor-start motor, if the starting torque is proportional to sin ϕ, what will happen if the phase angle between currents reduces from 90° to 60°? A) Torque increases by 33% B) Torque decreases by 33% C) Torque decreases by 50% D) Torque remains the same Answer. B. Torque decreases by 33%	3	Entered
35	A shaded-pole motor has shading coil resistance R_s=1Ω and inductive reactance X_s=2Ω. Find the phase shift between shaded and unshaded flux.	4	Entered

	A) 26.6° B) 45° C) 63.4° D) 90° Answer. C. 63.4°		
36	A capacitor-start motor draws a line current of 5 A and power factor of 0.75 at starting. Find the active power input. A) 800 W B) 900 W C) 750 W D) 600 W Answer. C. 750 W	4	Entered
37	A capacitor-start and run motor has a main winding impedance 20 Ω, auxiliary winding impedance 15 Ω, and capacitance 20 μF at 50 Hz. Find the capacitive reactance. A) 159 Ω B) 127 Ω C) 106 Ω D) 100 Ω Answer. B. 127 Ω	4	
38	A 4-pole, 50 Hz, squirrel cage induction motor runs at 1450 rpm. Find the slip. A) 2% B) 3% C) 4% D) 5% Answer. C. 4%	5	Entered
39	A 6-pole, 50 Hz induction motor runs at 960 rpm. Find the slip. A) 0.02 B) 0.03 C) 0.04 D) 0.05 Answer. C. 0.04	3	Entered
40	A 3-phase, 4-pole, 50 Hz induction motor runs at 1470 rpm. Find the frequency of rotor currents. A) 1.5 Hz B) 2 Hz C) 3 Hz D) 5 Hz Answer. B. 2 Hz	4	Entered
41	A 3-phase induction motor has a rotor copper loss of 150 W when the slip is 3%. Find the rotor input power. A) 4500 W B) 5000 W C) 4000 W D) 3500 W Answer. B. 5000 W	3	Entered

42	A 3-phase, 50 Hz, 8-pole induction motor has a speed of 720 rpm. Find its slip and rotor frequency. A) 2%, 1 Hz B) 3%, 1.5 Hz C) 4%, 2 Hz D) 5%, 2.5 Hz Answer. C. 4%, 2 Hz	4	Entered
43	A 3-phase induction motor takes 10 kW rotor input at slip = 0.05. Find rotor copper loss and mechanical power developed. A) 250 W, 9750 W B) 500 W, 9500 W C) 750 W, 9250 W D) 1000 W, 9000 W Answer. B. 500 W, 9500 W	4	Entered
44	A 3-phase induction motor develops an output power of 18 kW with rotor copper loss = 900 W and stator losses = 600 W. Find the input power. A) 18.9 kW B) 19.5 kW C) 19.8 kW D) 20.1 kW Answer. C. 19.8 kW	3	Entered
45	A 3-phase induction motor operates at 4% slip and rotor copper loss = 600 W. Find the air-gap power. A) 12 kW B) 15 kW C) 18 kW D) 20 kW Answer. B. 15 kW	5	Entered
46	A 3-phase, 4-pole, 50 Hz squirrel cage motor develops 20 N·m torque at 1450 rpm. Find the mechanical power developed. A) 2500 W B) 2900 W C) 3040 W D) 3200 W Answer. C. 3040 W	5	Entered
47	A 3-phase induction motor has rotor copper loss = 400 W, mechanical losses = 200 W, stator losses = 300 W, and input = 8 kW. Find efficiency. A) 85% B) 87.5% C) 90% D) 92% Answer. B. 87.5%	4	Entered
48	A 3-phase induction motor running at slip 5% draws input 15 kW, with stator losses 500 W and rotor mechanical losses 300 W. Find shaft output power.	4	Entered

	A) 13.3 kW B) 13.8 kW C) 14 kW D) 14.2 kW Answer. A. 13.3 kW		
49	A 3-phase, 50 Hz, 6-pole motor operates at 970 rpm. Find the rotor current frequency. A) 1 Hz B) 2 Hz C) 2.5 Hz D) 3 Hz Answer. B. 2 Hz	3	Entered
50	If the starting torque of a 3-phase induction motor is 2.5 times the full-load torque, and full-load torque is 40 N·m, find the starting torque. A) 60 N·m B) 80 N·m C) 100 N·m D) 120 N·m Answer.C. 100 N·m	3	
51	A 3-phase squirrel cage motor has an efficiency of 88% and draws input 10 kW. Find the shaft output power. A) 8700 W B) 8800 W C) 9000 W D) 9200 W Answer. A. 8700 W	4	Entered
52	A 3-phase, 4-pole, 400 V motor delivers 12 kW at 1450 rpm. Find the developed torque. A) 70 N·m B) 75 N·m C) 78.9 N·m D) 82 N·m Answer. C. 78.9 N·m	4	
53	A 3-phase induction motor develops mechanical power = 9.5 kW at slip 4%. Find rotor copper loss. A) 200 W B) 300 W C) 400 W D) 500 W Answer. D. 500 W	5	
54	For an Induction motor, R_2=0.4 Ω, X_2=2Ω. Find the slip at maximum torque. A) 0.1 B) 0.2 C) 0.3 D) 0.4 Answer. B. 0.2	5	

55	If a 3-phase induction motor has R_2=0.5Ω, X_2=2Ω and slip = 0.05, find the torque ratio (T/T_{max}). A) 0.20 B) 0.45 C) 0.62 D) 0.80 Answer. C. 0.62	5	
56	For a 50 Hz, 4-pole induction motor, synchronous speed = 1500 rpm. If torque is maximum at slip = 0.1, find the speed at which maximum torque occurs. A) 1400 rpm B) 1350 rpm C) 1300 rpm D) 1200 rpm Answer. A. 1400 rpm	5	
57	If stator voltage is reduced by 10%, the maximum torque changes by: A) decreases by 10% B) decreases by 19% C) decreases by 20% D) remains constant Answer. B. decreases by 19%	4	
58	A 3-phase induction motor develops full-load torque at slip 5%, and maximum torque occurs at slip 20%. Find the ratio T_FL/Tmax. A) 0.20 B) 0.25 C) 0.40 D) 0.50 Answer. C. 0.40	4	
59	For a motor with R_2=0.5Ω,X_2=2Ω,S=0.05, determine the ratio of actual torque to starting torque. A) 0.25 B) 0.50 C) 0.75 D) 1.0 Answer. B. 0.50	3	
60	A 3-phase motor develops maximum torque at slip 0.1 with torque = 40 N·m. Find the torque at slip 0.05. A) 15 N·m B) 20 N·m C) 30 N·m D) 35 N·m Answer. C. 30 N·m	4	
61	If torque at full load (slip = 0.04) is 60 N·m and slip at max torque = 0.15, find maximum torque.	5	

	A) 70 N·m B) 100 N·m C) 120 N·m D) 150 N·m Answer. C. 120 N·m		
62	At high slip, torque $\propto$? A) s B) 1/s C) s^2 D) $1/s^2$ Answer. B. 1/s	4	
63	A 4-pole, 50 Hz motor has s_{max}=0.15. Find speed at maximum torque. A) 1500 rpm B) 1400 rpm C) 1350 rpm D) 1275 rpm Answer. C. 1350 rpm	4	
64	The torque is maximum when: A) R_2=sX_2 B) s=R_2/X2 C) sX_2=R_2 D) Both B and C Answer. D. Both B and C	5	
65	A single-phase transformer has N_1 = 1000 turns, N_2 = 200 turns, and the primary voltage is 1000 V. Find the secondary voltage. A) 100 V B) 200 V C) 400 V D) 500 V Answer. B. 200 V	4	Entered
66	A 230/115 V transformer has 460 turns on the primary. Find the number of secondary turns. A) 230 B) 115 C) 230 × 460 / 115 D) 230 Answer. B. 115	4	Entered
67	A transformer with an efficiency of 98% delivers 1960 W output. Find its total losses. A) 20 W B) 30 W C) 40 W D) 50 W Answer. C. 40 W	5	Entered
68	The iron loss of a transformer is 120 W, and the full-load copper loss is 180 W. Find the efficiency at half load (cosφ = 1).	5	Entered

	A) 94.1% B) 96% C) 97.2% D) 98% Answer. B. 96%		
69	At what load will the maximum efficiency occur for the transformer in Q4? A) 0.4 full load B) 0.6 full load C) 0.8 full load D) 1.0 full load Answer. C. 0.8 full load	3	
70	If the primary current of a 1 kVA, 250/125 V transformer is 4 A, find the secondary current. A) 8 A B) 4 A C) 2 A D) 16 A Answer. A. 8 A	4	Entered
71	A transformer has 2% voltage regulation at full load, unity power factor. At 0.8 lagging pf, the regulation will be approximately: A) 2.4% B) 3.0% C) 3.5% D) 4.0% Answer. A. 2.4%	3	Entered
72	A 100 kVA, 11000/400 V transformer has 2% resistance drop and 4% reactance drop. Find the voltage regulation at 0.8 lagging PF. A) 3.2% B) 4.8% C) 5.0% D) 6.0% Answer. B. 4.8%	4	Entered
73	Find the voltage regulation at 0.8 leading PF for the same transformer. A) 4.8% B) 2.4% C) 0.8% D) 0.0% Answer. C. 0.8%	5	Entered
74	A transformer takes 0.5 A at 230 V on open circuit. Find the iron loss if the power factor is 0.3 lagging. A) 25 W B) 30 W	3	Entered

	C) 34.5 W D) 40 W Answer. C. 34.5 W		
75	A 200/400 V, 50 Hz transformer has 400 turns on the secondary. Find the number of primary turns. A) 200 B) 100 C) 150 D) 160 Answer. A. 200	4	Entered
76	Which of the following is the main energy storage component in an electric vehicle? a) Alternator b) Fuel tank c) Battery pack d) Supercharger Answer: c) Battery pack	3	Entered
77	The energy stored (E) in a 400 V, 85 kWh EV battery is approximately: a) 2.1×10^8 J b) 3.06×10^8 J c) 6.12×10^8 J d) 8.5×10^5 J Answer: c) 6.12×10^8 J	4	Entered
78	A 300 V battery supplies 100 A to the motor for 2 hours. The theoretical driving capacity is: a) 30 Ah b) 200 Ah c) 600 Ah d) 300 Ah Answer: d) 300 Ah	5	Entered
79	In an EV, the inverter converts: a) AC to DC b) DC to AC c) AC to AC d) DC to DC Answer: b) DC to AC	3	
80	Series connection of multiple battery cells in an EV pack: a) Increases current capacity b) Reduces total voltage c) Increases total voltage d) Reduces energy density Answer: c) Increases total voltage	3	

81	An EV battery has a C-rate of 2C and a capacity of 50 Ah. The maximum charging current is: a) 25 A b) 50 A c) 100 A d) 150 A Answer: c) 100 A	4	Entered
82	Which parameter defines battery power capability? a) Ampere-hours b) Watt-hours per kilogram c) Energy density d) Power density Answer: d) Power density	3	
83	For vehicle safety, the isolation resistance between the high-voltage battery and chassis should not be lower than: a) 5 Ω b) 50 Ω c) 500 kΩ d) 1 MΩ Answer: c) 500 kΩ	4	
84	What is the primary function of a Battery Management System (BMS) in an electric vehicle? a) Convert DC to AC b) Monitor and ensure battery safety and longevity c) Increase battery capacity d) Regulate motor speed Answer: b) Monitor and ensure battery safety and longevity	4	
85	If a Li-ion battery has a nominal voltage of 3.7 V and a capacity of 50 Ah, what is its energy capacity in Wh? a) 185 Wh b) 3.7 Wh c) 50 Wh d) 1850 Wh Answer: a) 185 Wh	5	Entered
86	Which parameter indicates the battery's ability to deliver power quickly, often expressed in watts per kilogram? a) Energy density b) Power density c) Capacity d) State of Charge Answer: b) Power density	4	
87	The process of limiting battery charging current to protect against overheating and overcharging is known as: a) Thermal runaway b) Regenerative braking	3	

	c) Charge management d) Pulse charging Answer: c) Charge management		
88	State of Charge (SoC) is best described as: a) The battery's current voltage b) The remaining charge percentage relative to full capacity c) The battery's temperature d) Total energy delivered from the battery Answer: b) The remaining charge percentage relative to full capacity	3	
89	A battery with 90% charging efficiency requires how much input energy to store 45 kWh of usable energy? a) 40.5 kWh b) 50 kWh c) 45 kWh d) 36 kWh Answer: b) 50 kWh	4	Entered
90	Which component in a BMS accurately measures the battery pack voltage and temperature? a) Current sensor b) Voltage and temperature sensors c) Inverter d) Charger Answer: b) Voltage and temperature sensors	3	
91	Which charging method is characterized by charging at a constant current followed by constant voltage? a) Trickle charging b) Constant voltage charging c) CC-CV (Constant Current-Constant Voltage) charging d) Pulse charging Answer: c) CC-CV (Constant Current-Constant Voltage) charging	3	Entered
92	If a battery pack consists of 96 cells in series, each with 3.7 V nominal voltage, what is the total nominal pack voltage? a) 355.2 V b) 259.2 V c) 96 V d) 3.7 V Answer: a) 355.2 V	4	Entered
93	In EVs, regenerative braking helps to: a) Increase battery degradation b) Convert kinetic energy into electrical energy to recharge the battery c) Stop the motor instantly d) Cool down the battery pack Answer: b) Convert kinetic energy into electrical energy to recharge the battery	3	Entered

94	The resistance of a platinum RTD at 0°C is **100 Ω**. If the temperature coefficient of resistance (α) is **0.00385 Ω/Ω/°C**, what will be its resistance at 100°C? **A)** 138.5 Ω **B)** 120 Ω **C)** 150 Ω **D)** 180 Ω **Answer:** A) 138.5 Ω	5	Entered
95	An RTD with α = 0.00392 and R_0 = 100 Ω has a measured resistance of 119.6 Ω. Find the temperature. **A)** 40°C **B)** 50°C **C)** 45°C **D)** 55°C **Answer:** B) 50°C	4	Entered
96	A platinum RTD reads 110 Ω at 25°C. If its nominal resistance is 100 Ω at 0°C, find the approximate temperature coefficient (α). **A)** 0.00385 **B)** 0.00400 **C)** 0.00360 **D)** 0.00450 **Answer:** A) 0.00385	4	Entered
97	If an RTD sensor is connected through a 2-wire configuration, which type of error is introduced? **A)** Offset error due to lead resistance **B)** Gain error **C)** Temperature coefficient error **D)** No error **Answer:** A) Offset error due to lead resistance	3	
98	A 3-wire RTD has two equal lead resistances of 1 Ω each. If the RTD resistance is 120 Ω, what is the approximate percentage error if it were connected as a 2-wire RTD? **A)** 0.83% **B)** 1.6% **C)** 1.0% **D)** 2.5% **Answer:** B) 1.6%	5	Entered
99	A platinum RTD is used to measure a temperature range of 0–300°C. If the RTD has R_0 = 100 Ω and α = 0.00385, find the resistance at 250°C. **A)** 192 Ω **B)** 196.25 Ω **C)** 198 Ω **D)** 200 Ω **Answer:** B) 196.25 Ω	4	Entered
100	Which of the following materials is commonly used in RTDs due to its high linearity and stability? **A)** Copper	3	

	B) Nickel **C)** Platinum **D)** Aluminum **Answer:** C) Platinum		
101	An RTD has a resistance of 145 Ω at 120°C. If R_0 = 100 Ω, find the average temperature coefficient (α). **A)** 0.0036 **B)** 0.00375 **C)** 0.0039 **D)** 0.0040 **Answer:** B) 0.00375	5	Entered
102	If the excitation current through a 100 Ω RTD is 5 mA, what is the self-heating power dissipated? **A)** 0.5 mW **B)** 1.0 mW **C)** 2.5 mW **D)** 5 mW **Answer:** C) 2.5 mW	5	Entered
103	If an RTD's resistance at 0°C is 100 Ω and the temperature coefficient of resistance (α) is 0.00385 Ω/Ω/°C, what is the resistance at 100°C? a) 138.5 Ω b) 138.0 Ω c) 140 Ω d) 155 Ω Answer: a	5	Entered
104	What is the typical temperature range of Platinum RTDs? a) -50°C to 150°C b) -200°C to 850°C c) 0°C to 500°C d) 100°C to 1000°C Answer: b	4	Entered
105	If an RTD has a resistance of 120 Ω at 50°C and 140 Ω at 100°C, what is the resistance at 0°C? a) 100 Ω b) 110 Ω c) 90 Ω d) 130 Ω Answer: a	5	Entered
106	Which RTD type provides the highest accuracy? a) 2-wire RTD b) 3-wire RTD c) 4-wire RTD d) Thin-film RTD Answer: c	4	
107	Calculate the temperature if an RTD (with R0=100ΩR0=100Ω, α=0.00385α=0.00385) shows a resistance of 115.5 Ω a) 40°C	5	Entered

	b) 50°C c) 60°C d) 70°C Answer: b		
108	Why do RTDs have a longer response time compared to thermocouples? a) Due to their larger size b) Due to higher thermal mass c) Due to slower electronic circuitry d) Due to being chemically inert Answer: b	3	
109	Which measurement device is best suited for precise temperature readings between -200 and 600°C? a) Thermocouple b) Thermistor c) RTD d) LVDT Answer: c	4	
110	An RTD has a resistance of 110 Ω at 25°C. What will be the resistance at 125°C? α=0.00385α=0.00385 a) 120 Ω b) 150 Ω c) 153.5 Ω d) 167 Ω Answer: c	5	Entered
111	Which RTD wire configuration completely compensates for wire resistance? a) 2-wire b) 3-wire c) 4-wire d) Thin film Answer: c	3	
112	An RTD sensor has a resistance of 200 Ω at 30°C and 220 Ω at 80°C. Estimate temperature coefficient, αα. a) 0.004 b) 0.0025 c) 0.0023 d) 0.0019 Answer: c	4	Entered
113	Calculate the resistance of an RTD at 70°C if at 0°C resistance is 100Ω and α=0.0039α=0.0039. a) 127.3 Ω b) 130.3 Ω c) 120.3 Ω d) 110.3 Ω Answer: a	5	Entered
114	The temperature coefficient of resistance αα for copper is approximately:	4	Entered

	a) 0.00427 b) 0.00385 c) 0.00600 d) 0.00718 Answer: a		
115	RTD self-heating arises due to: a) High excitation current b) Large wire diameter c) External heat sources d) Environmental humidity Answer: a	4	
116	Calculate the change in resistance for a Pt RTD from 0°C to 50°C with R0=100ΩR0=100Ω. a) 19.25 Ω b) 15.25 Ω c) 13.5 Ω d) 20 Ω Answer: a	5	Entered
117	Resistance of RTD increases approximately how much per °C? a) 0.0035 Ω per Ω per °C b) 0.00385 Ω per Ω per °C c) 0.002 Ω per Ω per °C d) 0.001 Ω per Ω per °C Answer: b	5	Entered
118	The standard nominal resistance of a Pt100 RTD is: a) 1000 Ω at 0°C b) 100 Ω at 0°C c) 10 Ω at 0°C d) 1 Ω at 0°C Answer: b	4	Entered
119	The Callendar-Van Dusen equation is used to: a) Calculate temperature from resistance in RTDs b) Correct wire resistance c) Measure temperature using thermocouples d) Calculate resistance of strain gauges Answer: a	4	
120	Thin-film RTDs typically have: a) Longer life but less accuracy than wire-wound RTDs b) Higher accuracy than wire-wound RTDs c) Better shock resistance d) Both a and c Answer: d	3	
121	An RTD shows resistance of 112.5 Ω at 25°C. If R0=100ΩR0=100Ω, calculate temperature tt. a) 32.5°C b) 25°C c) 45°C	5	Entered

	d) 30°C Answer: a		
122	The voltage generated by a thermocouple for a temperature difference of 100°C and Seebeck coefficient of 40 μV/°C is: a) 0.4 mV b) 4 mV c) 40 mV d) 400 mV Answer: b) 4 mV	5	Entered
123	A Type K thermocouple has an output of 5.2 mV. The temperature difference measured is approximately: a) 100°C b) 130°C c) 150°C d) 180°C Answer: c) 150°C	3	
124	The cold-junction compensation is necessary because: a) Thermocouple voltage is linear with temperature b) The reference junction is never at 0°C c) The Seebeck coefficient changes with temperature d) It prevents current flow in the thermocouple Answer: b	3	
125	If a thermocouple outputs 1 mV at 40°C, what will be its output at 100°C, assuming linearity and same Seebeck coefficient? a) 2.5 mV b) 2.0 mV c) 1.5 mV d) 3.0 mV Answer: b	4	
126	Calculate the temperature when a Type J thermocouple reads 7.8 mV, assuming the cold junction is at 0°C, and Seebeck coefficient is 50 μV/°C. a) 156°C b) 140°C c) 162°C d) 168°C Answer: a	4	
127	The voltage output for a nickel-chromium thermocouple at 550°C is approximately: a) 15 mV b) 17 mV c) 14 mV d) 12 mV Answer: b	5	
128	If the cold junction is at 25°C and thermocouple measures a voltage equivalent to 200°C at the hot junction, what is the corrected temperature reading? a) 175°C	5	

	b) 225°C c) 200°C d) 215°C Answer: b		
129	Thermocouples generate voltage based on: a) Ohm's Law b) Faraday's Law c) Seebeck Effect d) Peltier Effect Answer: c	3	
130	Which thermocouple type has the largest temperature range? a) Type J b) Type K c) Type T d) Type B Answer: d	3	
131	For a Type T thermocouple, the voltage output at 0°C is: a) 0 mV b) 10 mV c) -6 mV d) 20 mV Answer: a	4	
132	At what temperature does the thermocouple voltage reach zero? a) 100°C b) 0°C c) 25°C d) -50°C Answer: b	4	
133	Thermocouples are most suitable for: a) Low temperature and high precision applications b) High temperature applications c) Measuring small temperature ranges only d) Underwater temperature sensing only Answer: b	3	
134	Thermocouple voltage output for 0.5 mV corresponds roughly to what temperature difference? (Assume 40 μV/°C Seebeck coeff.) a) 12.5°C b) 50°C c) 0.8°C d) 125°C Answer: a	4	
135	The Seebeck coefficient for a thermocouple changes primarily with: a) Humidity b) Temperature c) Pressure d) Voltage Answer: b	3	

136	How does increasing wire length affect a thermocouple measurement? a) Increases voltage output b) Decreases voltage output c) No effect on voltage output d) Changes temperature measurement Answer: c	3	
137	The typical accuracy of a Type K thermocouple is: a) ±0.1°C b) ±1.5°C to ±2.5°C c) ±5°C d) ±10°C Answer: b	5	
138	The cold junction is maintained at a different temperature than 0°C. How can the temperature be accurately measured? a) Measure the voltage only at the hot junction b) Add cold junction temperature to the hot junction temperature reading c) Use cold junction compensation circuitry d) Ignore cold junction temperature Answer: c	3	
139	Calculate hot junction temperature if measured thermocouple voltage equals 5 mV and cold junction temperature is 30°C assuming linear behavior and Seebeck coefficient 40 µV/°C. a) 95°C b) 100°C c) 125°C d) 155°C Answer: c	4	
140	What is the primary principle on which an LVDT operates? a) Mutual inductance b) Self-inductance c) Capacitance change d) Resistance change Answer: a) Mutual inductance	4	
141	An LVDT has a sensitivity of 20 mV/mm. If the core displacement is 0.5 mm, what is the output voltage? a) 10 mV b) 20 mV c) 25 mV d) 40 mV Answer: a) 10 mV	5	
142	The output voltage of an LVDT at null position (zero displacement) is: a) Maximum b) Minimum but non-zero c) Zero	3	

	d) 50% of max output Answer: c) Zero		
143	Two secondary coils in LVDT are connected in: a) Series aiding b) Series opposing c) Parallel d) Series neutral Answer: b) Series opposing	3	
144	What happens to the output voltage of LVDT when the core moves toward one secondary coil? a) Output voltage increases with positive polarity b) Output voltage decreases and polarity reverses c) Output voltage remains constant d) Output voltage shows random fluctuations Answer: a) Output voltage increases with positive polarity	3	
145	If an LVDT produces 5 V output for 10 mm core displacement, what is its sensitivity in mV/mm? a) 20 mV/mm b) 0.5 mV/mm c) 500 mV/mm d) 50 mV/mm Answer: d) 50 mV/mm	5	
146	In an LVDT, the frequency of the excitation voltage is generally: a) 50 Hz b) 60 Hz c) 1 kHz to 10 kHz d) 1 MHz Answer: c) 1 kHz to 10 kHz	4	
147	Calculate the displacement if the LVDT voltage output is 8 mV and sensitivity is 8 mV/mm. a) 1 mm b) 0.5 mm c) 2 mm d) 4 mm Answer: a) 1 mm	4	
148	Which factor can affect the accuracy of an LVDT measurement? a) External magnetic fields b) Temperature variations c) Mechanical vibration d) All of the above Answer: d) All of the above	3	
149	Which of the following is an advantage of LVDT over other displacement transducers? a) High sensitivity and resolution b) No friction and wear c) Infinite mechanical life d) All of the above Answer: d) All of the above	3	

150	Which of the following best represents the function of a Battery Management System (BMS)? a) Converts AC to DC b) Ensures safe charging and discharging c) Regulates motor torque d) Generates tractive power Answer: b) Ensures safe charging and discharging	3	